

Question Paper Preview

Notations :

- 1.Options shown in **green** color and with  icon are correct.
- 2.Options shown in **red** color and with  icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

AI

Section Id :	640653124550
Number of Questions :	10
Section Marks :	25
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531859397 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536024860.  YES

6406536024861.  NO

Question Number : 2 Question Id : 6406531859398 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question

**ASK FOR PRINTED
GRAPH SHEETS**
8 PAGES TWO-SIDE PRINT

Options :

6406536024862. ✓ Printed graph sheets were provided on time.

6406536024863. ✗ Printed graph sheets were provided late.

6406536024864. ✗ Printed graph sheets were not provided.

6406536024865. ✓ I used the graph sheets.

6406536024866. ✗ I did not use graph sheets.

Question Id : 6406531859399 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (3 to 7)

Question Label : Comprehension

SEARCH

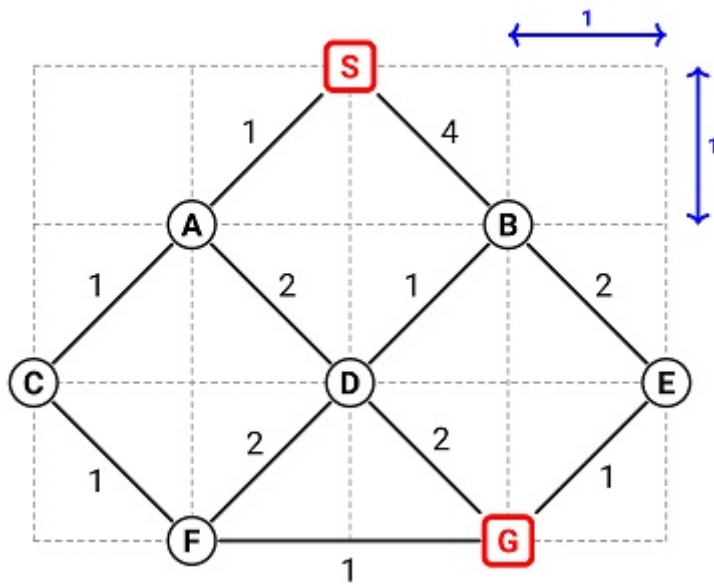
The figure shows a map on a uniform grid where each tile is 1x1 in size.

The start node is S and the goal node is G.

The MoveGen function returns nodes in **alphabetical** order.

Use Manhattan Distance as the heuristic function.

Tie-breaker: If several nodes have the same cost, use node labels to break the tie.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 3 Question Id : 6406531859400 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by the Best First Search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,B,D,G

S,B,E,G

Question Number : 4 Question Id : 6406531859401 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by A* search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,A,D,G

Question Number : 5 Question Id : 6406531859402 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Branch-and-Bound search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,A,C,F,G

Question Number : 6 Question Id : 6406531859403 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the given map, which algorithm finds the shortest path from S to G?

Options :

6406536024870. ✖ A* Search Algorithm

6406536024871. ✔ Branch-and-Bound Search Algorithm

Both A* and Branch-and-Bound Search Algorithms

6406536024872. ✖

6406536024873. ✖ None of these

Question Number : 7 Question Id : 6406531859404 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Select the correct statement about the given graph.

Options :

6406536024874. ✖ Heuristic is admissible.

6406536024875. ✔ Heuristic is not admissible.

6406536024876. ✖ Heuristic is admissible in some cases and not admissible in other cases.

6406536024877. ✖ There is not enough information to determine admissibility.

Question Id : 6406531859405 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (8 to 9)

Question Label : Comprehension

GAMES: ALPHA-BETA

Consider a game tree with the root node as MAX.

Alpha-Beta algorithm is in mid-flight currently processing a path from the root to a node labeled N.

The snapshot of evals of nodes along the current path is: 2, 14, 4, 10, N.

Where evals are restricted to EVEN NUMBERS greater than zero and less than 15.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 8 Question Id : 6406531859406 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Determine the eval of node N that will induce a cutoff to prune its siblings.

Enter an even number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 16

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

2

4

Question Number : 9 Question Id : 6406531859407 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What type of cut-off is induced by the eval selected for N?

Options :

6406536024879. ✓ Alpha Cutoff

6406536024880. ✗ Beta Cutoff

6406536024881. ✗ Alpha-Beta Cutoff

6406536024882. ✗ None of these

Question Id : 6406531859408 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (10 to 11)

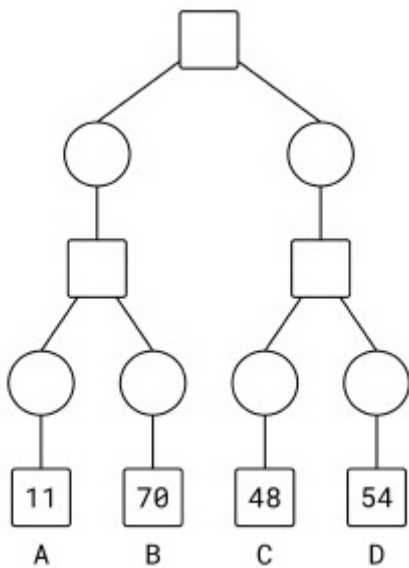
Question Label : Comprehension

GAMES: SSS STAR

The figure shows a game tree with evaluation function values at the horizon nodes. The horizon nodes are labeled from A to D. Use these labels to enter a horizon node or a list of horizon nodes in short answers.

Tie-breaker:

When several nodes carry the same best cost then select the deepest node, if tie persists then select the leftmost of the deepest nodes to break the tie.



Run SSS* algorithm on the game tree, then answer the sub-questions.

Sub questions

Question Number : 10 Question Id : 6406531859409 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes assigned SOLVED status by SSS* algorithm, then enter the labels of those nodes in the textbox, or enter NIL if no nodes are assigned SOLVED status.

Enter node labels as a comma separated list in **alphabetical** order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,C,D

Question Number : 11 **Question Id :** 6406531859410 **Question Type :** SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes pruned by SSS* algorithm,
i.e., horizon nodes removed from the queue by

MAX-ancestor. Enter the labels of those nodes
in the textbox, or enter NIL if no nodes are pruned.

Enter node labels as a comma separated list in
alphabetical order.

NO SPACES, TABS, DOTS, BRACKETS OR
EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,C,D

Question Id : 6406531859411 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (12 to 14)

Question Label : Comprehension

PROBLEM DECOMPOSITION

The figure shows an AND-OR decomposition of problem S into smaller problems. The nodes are uniquely identified by labels (S, A, B, C, ...).

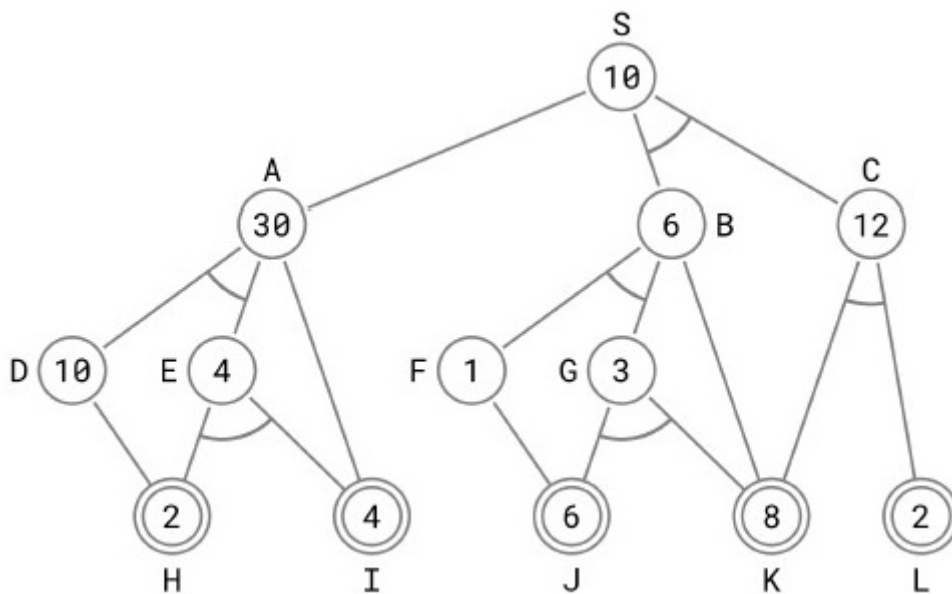
Each node shows the heuristic estimate of the cost of solving that node. Nodes shown in double lines are primitive nodes and their values are actual costs.

A primitive node is added to the graph, with SOLVED status, when its parent is expanded. And therefore, a primitive node is never expanded.

The cost of each edge is 2 units.

Tie-breaker 1: If several nodes have the same cost then break the tie using node labels.

Tie-breaker 2: For AND nodes, select the unsolved branch with the highest cost.



Use AO* algorithm to solve S, then answer the sub-questions.

Sub questions

Question Number : 12 Question Id : 6406531859412 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the nodes expanded by AO* algorithm.

List the nodes in the order they are expanded. Observe that primitive nodes are not expanded.

Enter a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS
OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,C,B,G

C,B,G

Question Number : 13 Question Id : 6406531859413 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

For each node expanded by AO* algorithm,
determine the value propagated to the
start node S. Enter the values of S as a list.

Enter a comma separated list of numbers.

NO SPACES, TABS, DOTS, BRACKETS
OR EXTRANEIOUS CHARACTERS.

Answer format: 12,42,17

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

22,24,26,28

24,26,28

Question Number : 14 Question Id : 6406531859414 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What can you conclude about the given AND-OR decomposition?

Options :

6406536024887. ✖ The heuristic is admissible.

6406536024888. ✔ The heuristic is inadmissible.

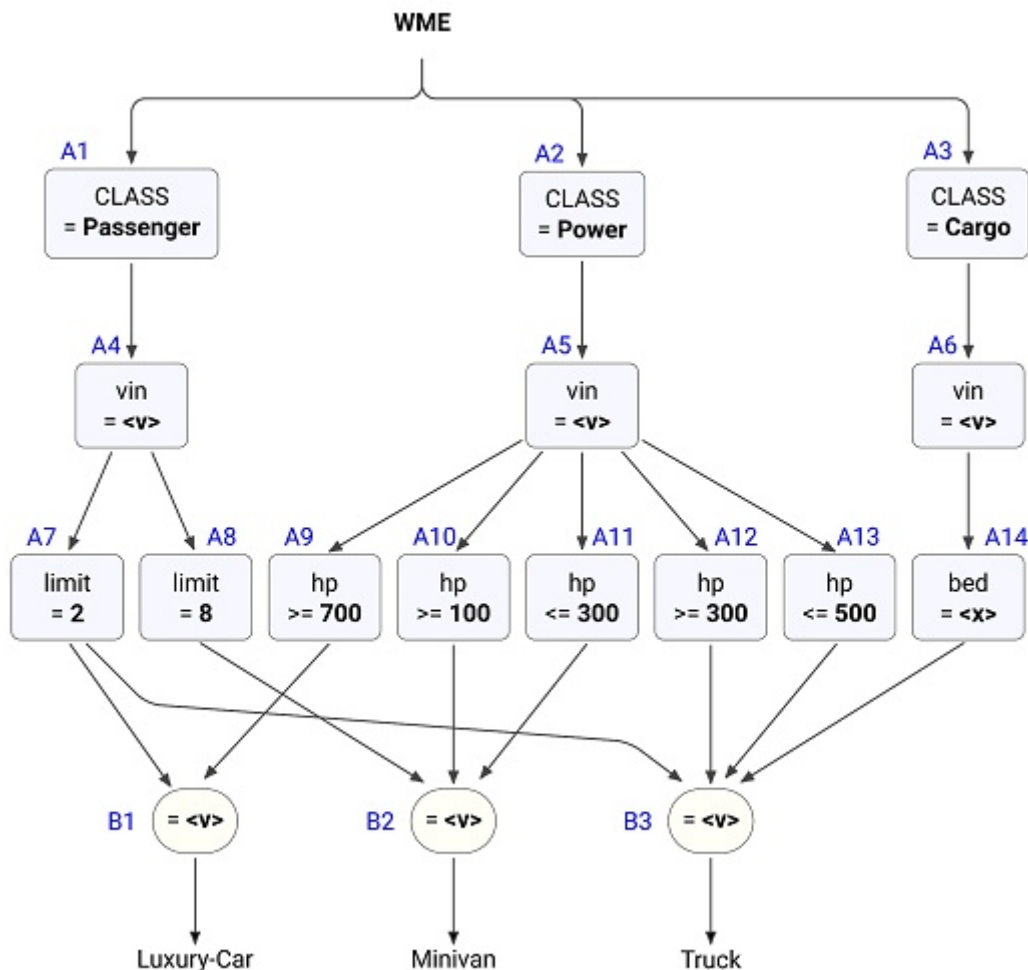
6406536024889. ✖ The heuristic is sometimes admissible and sometimes inadmissible.

Question Id : 6406531859415 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (15 to 17)

Question Label : Comprehension

RULE BASED EXPERT SYSTEMS

A Rete Net for classification of vehicles is shown in the figure. Labels A1, A2, A3, ..., A10, A11, A12, A13, ..., and B1, B2, B3 uniquely identify nodes in the network. When required, use the above label ordering to **break ties** and to enter short answers.



The Working Memory contains the following WMEs uniquely identified by timestamps (sequence numbers). Assume that WMEs reside in the appropriate Alpha node, and Beta nodes simply point to WMEs in the Alpha nodes.

- 101. (Cargo ^vin A1 ^bed flat)
- 102. (Passenger ^vin A1 ^limit 2)
- 103. (Passenger ^vin B2 ^limit 8)
- 104. (Passenger ^vin C3 ^limit 2)
- 105. (Power ^vin A1 ^hp 400)
- 106. (Power ^vin B2 ^hp 250)
- 107. (Power ^vin C3 ^hp 900)
- 108. (Wheel ^vin B2 ^tyre regular)
- 109. (Wheel ^vin C3 ^tyre regular)

For each WME identify its location (node label) in the Rete Net, then prepare the conflict set for the first Match-Resolve-Execute cycle, then answer the sub-questions.

Sub questions

Question Number : 15 Question Id : 6406531859416 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following rule-data tuples occur in the conflict set?

Options :

6406536024890. ✓ Luxury-Car,104,107

6406536024891. ✓ Minivan,103,106

6406536024892. ✓ Truck,101,102,105

6406536024893. ✗ Minivan,103,106,108

Question Number : 16 Question Id : 6406531859417 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Specificity** for conflict resolution then which rule-data tuple will fire in the first round?

Options :

6406536024894. ✗ Luxury-Car,104,107

6406536024895. ✗ Minivan,103,106

6406536024896. ✓ Truck,101,102,105

6406536024897. ✗ Minivan,103,106,108

Question Number : 17 Question Id : 6406531859418 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Recency** for conflict resolution which rule-data tuple will fire in the first round?

Options :

6406536024898. ✓ Luxury-Car,104,107

6406536024899. ✖ Minivan,103,106

6406536024900. ✖ Truck,101,102,105

6406536024901. ✖ Minivan,103,106,108

Question Id : 6406531859419 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (18 to 20)

Question Label : Comprehension

AUTOMATED PLANNING 1

Answer the given subquestions.

Sub questions

Question Number : 18 Question Id : 6406531859420 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Consider actions a and b and two **feasible** orderings (a then b) and (b then a). Which of the following conditions will produce different outcomes for each ordering?

Options :

6406536024902. ✖ P is in pre(a) and also in del-effects(b).

6406536024903. ✔ P is in add-effects(a) and also in del-effects(b).

6406536024904. ✖ None of these.

Question Number : 19 Question Id : 6406531859421 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In planning graphs constructed by GraphPlan, actions a and b in layer n are mutex _____ .

Options :

6406536024905. ✔ if P in pre(a) and Q in pre(b) are mutex

6406536024906. ✔ if P is in pre(a), in del-effects(a), in pre(b) and also in del-effects(b)

6406536024907. ✔ if P is in pre(a) and also in del-effects(b)

6406536024908. ✔ if P is in add-effects(a) and also in del-effects(b)

Question Number : 20 Question Id : 6406531859422 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In planning graphs constructed by GraphPlan, propositions P and Q in layer n are mutex _____ .

Options :

6406536024909. ✔ if every action pair (a,b) with P in add-effects(a) and Q in add-effects(b) in layer n is mutex

6406536024910. ✖ if at least one action pair (a,b) with P in add-effects(a) and Q in add-effects(b) in layer n is mutex

6406536024911. ✖ none of these

Question Id : 6406531859423 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (21 to 24)

Question Label : Comprehension

AUTOMATED PLANNING 2

The domain description of a Blocks World with a single one-armed robot is given below.

PREDICATES

<code>armEmpty</code>	The arm is not holding any block, it is empty.
<code>holding(X)</code>	The arm is holding X.
<code>onTable(X)</code>	X is on the table.
<code>clear(X)</code>	X has nothing above it, it is clear.
<code>on(X,Y)</code>	X is directly placed on Y.

OPERATORS

`Pickup(X)`: pick up X from the table.

Preconditions: { `armEmpty`, `clear(X)`, `onTable(X)` }

Add Effects : { `holding(X)` }

Del Effects : { `armEmpty`, `onTable(X)` }

`Putdown(X)`: place X on the table.

Preconditions: { `holding(X)` }

Add Effects : { `armEmpty`, `onTable(X)` }

Del Effects : { `holding(X)` }

`Unstack(X,Y)`: pick up X that is directly sitting on Y.

Preconditions: { `armEmpty`, `clear(X)`, `on(X,Y)` }

Add Effects : { `clear(Y)`, `holding(X)` }

Del Effects : { `armempty`, `on(X,Y)` }

`Stack(X,Y)`: place X directly on top of Y.

Preconditions: { `holding(X)`, `clear(Y)` }

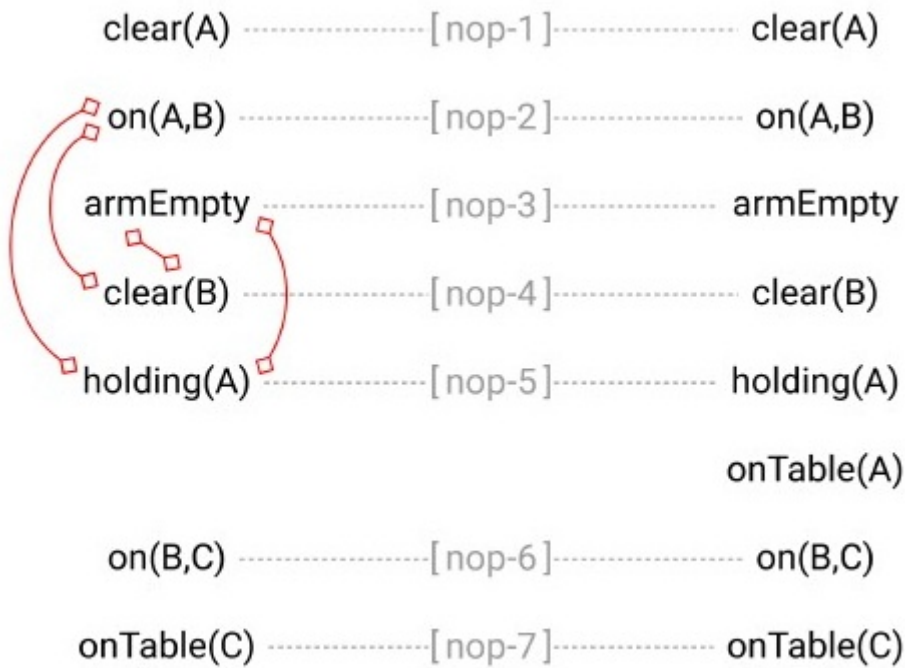
Add Effects : { `armEmpty`, `on(X,Y)` }

Del Effects : { `holding(X)`, `clear(Y)` }

NOTE: variables X and Y must bind to different objects

The GraphPlan algorithm is in mid-flight solving a planning problem, from the planning graph two consecutive propositional layers (layer k and k+1) are presented in the figure.

Both proposition layers are fully populated (no missing propositions).



Mutex proposition pairs in layer k are:

(on(A,B), holding(A)), (on(A,B), clear(B)), (armEmpty, clear(B)), (armEmpty, holding(A)).

Populate the action layer and mutex links, then answer the sub-questions.

Sub questions

Question Number : 21 Question Id : 6406531859424 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What can you conclude about layer k?

Options :

6406536024912. ✖ Layer k describes the start state of the planning problem.

6406536024913. ✔ Layer k cannot describe the start state of the planning problem.

6406536024914. ✖ There is insufficient information to comment on layer k.

Question Number : 22 Question Id : 6406531859425 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions in layer k+1?

Options :

6406536024915. ✔ Putdown(A)

6406536024916. ✔ Stack(A,B)

6406536024917. ✔ Unstack(A,B)

6406536024918. ✖ Unstack(B,C)

Question Number : 23 Question Id : 6406531859426 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **mutex** pairs in layer k+1?

Options :

6406536024919. ✓ Putdown(A) and Stack(A,B)

6406536024920. ✓ Stack(A,B) and Unstack(A,B)

6406536024921. ✓ nop-3 and nop-4

Question Number : 24 Question Id : 6406531859427 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

armEmpty and clear(B) in layer k+1 are non mutex because _____ .

Options :

6406536024922. ✗ nop-3 and nop-4 are non mutex

6406536024923. ✓ PutDown(A) and nop-4 are non mutex

6406536024924. ✗ None of these

Question Id : 6406531859428 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (25 to 27)

Question Label : Comprehension

CONSTRAINT SATISFACTION

Consider a CSP over 3 variables A, B, C (processed in that order) with domains and constraints as shown below.

$$D_A = D_B = D_C = \{1, 2, \dots, 9\}$$

$$R_{AB} = \{ (2, 6), (3, 5) \}$$

$$R_{AC} = \{ (2, 7), (2, 8) \}$$

$$R_{BC} = \{ (4, 7), (5, 7), (6, 8) \}$$

Draw the constraint graph and matching-diagram then answer the sub-questions.

Sub questions

Question Number : 25 Question Id : 6406531859429 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the given CSP arc consistent?

Options :

6406536024925. ✖ Yes

6406536024926. ✔ No

6406536024927. ✖ Cannot be determined

Question Number : 26 Question Id : 6406531859430 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the given CSP is not already arc consistent, make it arc consistent, then check if it is path consistent.

Options :

6406536024928. ✔ It is path consistent.

6406536024929. ✖ It is not path consistent.

6406536024930. ✖ Path consistency does not apply because the constraint graph is cyclic.

Question Number : 27 Question Id : 6406531859431 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Does the given CSP have a solution?

Enter the solution for the variables A,B,C as a comma separated list.
Enter NIL if there is no solution.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 1,2,3

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

2,6,8